

Section II

The Science and Physiology of Alternative Techniques for Increasing Range of Motion



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12 Effects of Resistance Training on Range of Motion

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The historical roots of strength training can be traced back to ancient civilizations, with the earliest recorded instances dating back to the Chou dynasty in China around 1122–249 BCE (10). However, a more structured approach to strength training emerged during the 6th century BCE with the legend of Milo of Croton, a Greek wrestler. Milo's remarkable athletic accomplishments, which included an impressive tally of 32 wrestling victories, were purportedly influenced by his unorthodox approach to building strength. According to legend, it was said that Milo would carry a young calf on his shoulders every day. As the calf grew into a fully grown and heavier animal over time, so too did Milo's strength increase proportionally. While some may question the historical accuracy surrounding this account, its significance lies in its symbolic representation of progressive resistance training – an essential concept revolving around systematically increasing one's workload with the aim of bolstering muscle strength, power, and endurance.

Milo's tale acts as an early precursor in history for our modern comprehension and appreciation of strength training methodologies centred around progressive overload principles. Despite being anecdotal in nature, this story underscores ancient insights recognizing the association between resistance-based exercises and their potential impact on augmenting overall muscular strength – ultimately providing a fundamental groundwork upon which further advancements within exercise science and physical conditioning could be built upon.

The origins of research on resistance training can be traced back to the late 1800s. For example, the work conducted by Scripture and colleagues in 1894 (29), regarding the cross-education effect of strength training represented an early exploration into a phenomenon where individuals experienced improvements not only in their trained limb but also in their untrained limb. This discovery suggested that strength adaptations were not limited to specific limbs but had systemic or at least contralateral implications, which served as a basis for future investigations into the broader effects of resistance training as well as more recently; global effects of stretching (see Chapter 6).

However, it was during the mid-1900s that resistance training research underwent its most significant paradigm shift. This period witnessed a surge of diverse studies focusing on different aspects of resistance training. Scholars

and scientists devoted themselves to comprehending and optimizing various elements (e.g., resistance training volume, intensity, frequency, recovery periods, movement velocity and others) associated with this type of exercise, to enhance strength, power output, and deciphering the physiological mechanisms underlying muscle hypertrophy.

However, even in the 1970s resistance training was viewed with scepticism as it was believed to lead to muscle-boundness. Muscle-boundness referred to more stiff, hypertrophied muscles, which impaired muscle elasticity (flexibility), speed, and coordination. Hence, resistance training was not recommended for most athletes. In 1964 J.R. Leighton's research shed light on an unexpected aspect of resistance training (19). While the primary focus of such training had traditionally been on augmenting physical strength, his study revealed that individuals engaging in resistance training not only experienced increases in strength but also demonstrated a significant enhancement in their range of motion (ROM). Leighton's work marked a paradigm shift, challenging the conventional view of resistance training as a unidimensional approach solely geared towards building muscular strength and size. The acknowledgment of improved ROM as an unintended yet consequential outcome broadened the scope of understanding surrounding the physiological adaptations associated with resistance training.

This revelation underscored the holistic impact of resistance training on human movement and functional capacity, emphasizing that its benefits extended beyond mere muscular development. By recognizing the interplay between strength gains and enhanced flexibility, Leighton's research contributed to a more nuanced appreciation of the multifaceted advantages associated with resistance training in the context of comprehensive physical fitness. This recognition, rooted in empirical evidence, laid the groundwork for subsequent scholarly inquiries into the diverse physiological adaptations resulting from resistance training.

Moderating Variables

When examining how strength training affects flexibility (ROM), it's important to distinguish between different types of resistance exercises. The primary isoinertial (resistance training that maintains a constant inertia throughout the ROM, facilitating a relatively constant resistance at every angle) types use free weights, machines, resistance bands, and body weight (e.g., calisthenics). Each of these methods has its own merits, but not all will improve the ROM of an individual (1).

Our meta-analysis showed that all types of strength training, except for using just your body weight (e.g., calisthenics such as push-ups), can increase ROM (1). When considering the mechanics of free weights, machines, and Pilates resistance training, one might suggest that these actions are similar to dynamic stretching, albeit with an additional external load (1). Dynamic stretching is an action that involves a controlled movement through the active joint ROM (3, 4, 9). Free weights, machines, and resistance bands resistance

training typically allows the joints to reach their endpoint ROM or the individual's point of initial or maximum discomfort at a controlled pace. Remember, research shows that it is unnecessary to go to the maximum ROM or discomfort to achieve gains in ROM (2, 3, 6). Hence, it is also unnecessary to move heavy weights to an extreme ROM where according to the force-length relationship of muscle (8, 14, 21), it is at its weakest point and thus would be more susceptible to injury.

In contrast, resistance training with body mass activities may not always permit such an expansive ROM. For example, while a push-up is restricted by chest circumference and the surface (i.e., floor or ground), free weights and machines can permit the shoulder to surpass this more restricted ROM. There is a dearth of studies comparing the effects of partial versus full ROM resistance training on ROM. Kawama et al. (16) reported significantly greater decreases in the shear modulus (muscle stiffness) of the semimembranosus (hamstrings muscle) with eccentric resistance training through a wide ROM versus either eccentric resistance training through a narrow ROM or concentric resistance training with a wide ROM. Furthermore, dynamic stretching involves repeated cyclical muscle loading and unloading (9). The addition of an external load with RT would augment the stress on musculotendinous and connective tissue and minimize the unloading component. As there are no studies comparing dynamic stretch training and resistance training on ROM, possible differential effects of these two activities on ROM should be investigated.

When examining additional factors that may impact improvements in ROM through resistance training, the training status of individuals emerges as a significant moderating variable. Notably, there is a discernible influence on outcomes based on whether individuals are categorized as sedentary or active. Surprisingly, sedentary individuals have exhibited greater improvements in ROM with resistance training compared to their active counterparts (1). This discrepancy can be linked to the baseline level of flexibility, which sets the stage for the greater magnitude of training-induced ROM enhancements. The observed trend can be explained by considering the pre-existing conditions of musculotendinous units. Sedentary individuals, starting with a lower baseline of flexibility, appear to experience more substantial gains in ROM as a result of resistance training. In contrast, individuals with an active lifestyle, particularly those engaged in regular dynamic loading exercises, may already have an elevated baseline of flexibility. Consequently, their potential for training-induced ROM increases could be somewhat muted compared to those who were previously untrained (20). It is noteworthy, however, that even individuals with a trained background still undergo significant improvements in ROM, albeit to a lesser degree than their untrained counterparts. This emphasizes the adaptability of the human body and the continued potential for enhanced flexibility through targeted resistance training, even for those who have a history of regular physical activity. Understanding how training status interacts

with the outcomes of resistance training on ROM is pivotal for tailoring effective training programmes based on individual characteristics and goals.

It is surprising that factors such as training frequency, training duration, age, and sex do not appear to significantly influence the outcomes of improvements in ROM through resistance training (1). Furthermore, it has been observed that the magnitude of ROM improvements does not exhibit notable variations across different joints (1). This finding implies that the benefits of resistance training for ROM are relatively consistent across various joints. Unlike some fitness outcomes that might show joint-specific responses, the impact of resistance training on flexibility appears to be more uniformly distributed.

Understanding these moderating variables provides valuable insights for designing comprehensive resistance training programmes aimed at improving flexibility. The lack of pronounced effects based on training frequency and sex suggests that individuals, regardless of their sex or how often they engage in resistance training, can experience comparable gains in ROM. Moreover, the uniformity in ROM improvements across joints underscores the holistic nature of resistance training's impact on flexibility throughout the body.

Possible Mechanisms

Various theories have been proposed to explain why resistance training improves ROM. In the context of neural adaptations associated with stretching and resistance training, particularly isoinertial resistance training, researchers have explored various mechanisms. Reports suggest that static stretch training over three to six weeks reduces tonic Ia afferent feedback from muscle spindles, which affects T-reflexes and H-reflexes (7, 11). This reduction in afferent feedback could potentially lead to diminished reflex-induced contractions, resulting in a more relaxed muscle state, referred to as disfacilitation. However, it's worth noting that dynamic stretching and isoinertial resistance training are more likely to excite, rather than disfacilitate, muscle spindle activity. While Golgi tendon organ inhibition is plausible with large amplitude stretches and higher muscle tension (12), it tends to subside immediately after the stimulus discontinues, making it an unlikely candidate for chronic dynamic stretching or isoinertial resistance training mechanisms.

Recurrent or Renshaw cell inhibition, which is associated more with acute dynamic rather than tonic contractions (15), may induce stabilizing effects on motoneuron discharge variability and motor unit synchronization (27). However, it's unclear whether these acute neural responses lead to chronic training adaptations.

Morphologically, dynamic ballistic stretch training has shown some evidence of decreasing muscle and tendon stiffness (25). Acute dynamic stretching has been associated with decreased passive resistive torque (17) and muscle stiffness (13, 17), suggesting a more compliant musculotendinous unit following a single session. However, a six-week ballistic stretch training programme did not reveal significant changes in muscle morphology (18).

Furthermore, there's a debate regarding the effects of resistance training on tendon and muscle stiffness. Some suggest that resistance training increases tendon stiffness through modifications in muscle elasticity (30), while others argue for an increase in muscle stiffness with resistance training (22). Eccentric actions with resistance training, which involve force production at extended positions, may increase ROM by altering fascicle length (26, 28) and pennation angle (28).

Notably, there is evidence supporting the increase in stretch (pain) tolerance with stretching (5, 23, 24) and a similar increase in pain tolerance is speculated with isoinertial resistance training, owing to the discomfort associated with external torques on muscles and joints. The combined impact of changes in musculotendinous unit stiffness and compliance, altered stretch tolerance, and potential adjustments in fascicle length and pennation angle may collectively contribute to the observed increase in ROM with dynamic stretching under load, such as isoinertial resistance training.

Summary

The evidence suggests that resistance training with external loads alone has the potential to improve ROM, making additional stretching either before or after resistance training potentially unnecessary for enhancing flexibility. Current studies and literature support the idea that both stretching and resistance training independently contribute to improvements in ROM, strength enhancement, and a reduction in the incidence of musculotendinous injuries.

In situations where time constraints or specific circumstances dictate, the benefits of flexibility training can be integrated into resistance training routines. However, it's important to recognize that stretch training still holds its place as a valuable component in overall fitness and training for a broad population. For instance, in scenarios where resistance training might not be suitable as part of a warm-up before competition, stretching can play a pivotal role in preparing the body for specific activities or competitions. Moreover, stretching serves as a relaxation method for many practitioners, where the dynamic and sometimes more intense nature of resistance training might not be as appropriate for achieving a state of relaxation.

While resistance training alone can positively impact flexibility, there are situations where incorporating dedicated stretch training remains beneficial and relevant. The choice between stretching and resistance training can be context-dependent, taking into consideration individual preferences, time constraints, and specific fitness or competition requirements. Both modalities, when appropriately applied, contribute to a well-rounded approach to physical preparation and maintenance. Illustrations of various foam rolling exercises can be found with Figures 12.1–12.13.



Figure 12.1 Dumbbell chest press / flies with arms extended



Figure 12.2 Dumbbell chest press / flies (horizontal abduction)



Figure 12.3 Pullover with initial position



Figure 12.4 Pullover with full shoulder flexion



Figure 12.5 Overhead triceps extensions



Figure 12.6 Lat pulldown with initial position (arms extended)



Figure 12.7 Lat pulldown with final position (shoulder horizontal extension)



Figure 12.8 Seated rowing with elbows extended and scapula protracted



Figure 12.9 Seated rowing with elbows flexed and scapula retracted



Figure 12.10 Incline bicep curls with elbows extended and shoulders in extension



Figure 12.11 Incline bicep curls with elbows flexed



Figure 12.12 Forward lunge



Figure 12.13 Step-up

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